

Small Open Economy Model

1. Model

We consider a representative agent in a small open economy who maximizes lifetime utility over consumption starting at $t = 0$:

$$\max_{\{c_t, s_{t+1}\}_{t=0}^{T-1}} \sum_{t=0}^{T-1} \beta^t \ln(c_t) + \phi \ln(1 - \ell_t) \quad (1)$$

subject to the following constraints:

1.1 Production

$$y_t = \theta_t l_t \quad (2)$$

1.2 Wages

$$w_t = \theta_t \quad (3)$$

1.3 Labor Supply (static intratemporal FOC)

$$l_t = 1 - \phi \frac{c_t}{w_t} \quad (4)$$

1.4 Current Account Constraint

Treating foreign assets as a decision variable, the current account constraint links consumption and asset accumulation:

$$CA_t = (s_{t+1} - s_t) = y_t - c_t + r_t s_t, \quad t = 0, \dots, T-1 \quad (5)$$

1.5 Initial Assets

$$s_0 \text{ given} \quad (6)$$

1.6 Euler Equation (Consumption FOC)

$$c_{t+1} = \beta(1 + r_{t+1}^*)c_t, \quad t = 0, \dots, T-2 \quad (7)$$

Alternatively: Endogenous Interest Rate (Risk Premium)

The effective interest rate faced by the economy depends negatively on its net foreign asset position:

$$r_t = r_t^w - \chi s_t \quad (8)$$

where $\chi > 0$ measures the sensitivity of the risk premium to external debt.

A negative foreign asset position ($s_t < 0$) increases the domestic interest rate through the risk premium term χs_t . This amplifies the cost of external borrowing and limits debt accumulation, ensuring model stability.

2. Steady-State

At the steady state ($c^*, s^*, ca^*, l^*, y^*, w^*$), variables are constant:

$$s^* : \quad ca^* = 0 \quad (9)$$

$$r^* : \quad r^* = \frac{1}{\beta} - 1 \quad (10)$$

$$c^* : \quad c^* = \frac{\theta}{1 + \phi} + \frac{1}{1 + \phi} \cdot \frac{1 - \beta}{\beta} s_0 \quad (11)$$

$$l^* : \quad l^* = 1 - \phi \frac{c^*}{w^*} \quad (12)$$

$$y^* : \quad y^* = \theta l^* \quad (13)$$

$$w^* : \quad w^* = \theta \quad (14)$$

Steady-State Consumption

From the labor supply condition,

$$\ell^{ss} = 1 - \phi \frac{c^{ss}}{w^{ss}}, \quad w^{ss} = \theta,$$

steady-state output is

$$y^{ss} = \theta \ell^{ss} = \theta - \phi c^{ss}.$$

Imposing the steady-state current account condition $CA^{ss} = 0$,

$$0 = y^{ss} - c^{ss} + r s_0 = \theta - \phi c^{ss} - c^{ss} + r s_0 = \theta - (1 + \phi) c^{ss} + r s_0,$$

and solving for c^{ss} gives

$$c^{ss} = \frac{\theta}{1 + \phi} + \frac{1}{1 + \phi} \cdot r s_0. \quad (15)$$

Using the steady-state Euler condition for log utility, $r = \frac{1 - \beta}{\beta}$, we finally obtain

$$c^{ss} = \frac{\theta}{1 + \phi} + \frac{1}{1 + \phi} \cdot \frac{1 - \beta}{\beta} s_0. \quad (16)$$

Steady-State Foreign Assets with no endogenous risk prime

The current account identity is given by:

$$CA_t = y_t - c_t + r^* s_t. \quad (17)$$

In the steady state, the current account must be zero, $CA^* = 0$, which implies:

$$0 = y^* - c^* + r^* s^*. \quad (18)$$

Solving for the steady-state level of foreign assets yields:

$$s^* = \frac{c^* - y^*}{r^*}. \quad (19)$$

Steady-State Foreign Assets with endogenous risk prime

From the steady-state Euler equation,

$$1 = \beta(1 + r^*) \quad \Rightarrow \quad r^* = \frac{1}{\beta} - 1, \quad (20)$$

and the interest rate rule with endogenous risk premium,

$$r^* = r^w - \chi s^*, \quad (21)$$

we obtain the steady-state level of foreign assets:

$$s^* = \frac{r^w - \left(\frac{1}{\beta} - 1\right)}{\chi} = \frac{r^w + 1 - \frac{1}{\beta}}{\chi}. \quad (22)$$

A higher sensitivity parameter χ implies a smaller deviation of s^* from zero for a given interest rate differential.